

MKN1 – MKN1 TASK 2: NON-RELATIONAL DATABASE DESIGN AND IMPLEMENTATION

DATA MANAGEMENT – D597

PRFA – MKN1

Preparation

Task Overview

Submissions

Evaluation Report

COMPETENCIES

4157.1.1 : Recommends an Appropriate Data Architecture

The learner recommends an appropriate data architecture.

4157.1.2 : Examines the Data Available for Analysis

The learner examines the data available for analysis to determine their dimension, quality, relations, and limitations.

4157.1.4 : Implements Physical Data Models

The learner implements physical data models by incorporating required data.

4157.1.5 : Performs Queries to Answer a Business Question

The learner performs database queries to answer a business question.

4157.1.6 : Uses Appropriate Data Manipulation Language Statements

The learner uses appropriate data manipulation language statements to manage data.

INTRODUCTION

A common undertaking for a data analyst is to create databases using existing datasets. This process involves the exploration of the source and target datasets to merge the data in a meaningful and logical way.

You will replicate this common process as you complete this task. You will take external data from either a document dataset or a graphical dataset and design a non-relational database solution. You will load the data into a database solution and retrieve or modify the data using JSON datasets and MongoDB in the virtual lab environment.

Choose one of the scenarios provided in the Supporting Documents. Each scenario has associated JSON files that will be located on the desktop of the WGU Virtual Lab environment. You will complete the following:

1. Design a **non-relational** database solution.
2. Implement the solution.
3. Present the solution using Panopto to an audience that consists of your peers within a project team.

Note: It is recommended to use the other scenario, different from the one you used in Task 1.

You will complete this performance assessment in the provided WGU Virtual Lab environment. Your submission will be a design document that includes screenshots of your work in the lab environment and a Panopto video.

REQUIREMENTS

Your submission must be your original work. No more than a combined total of 30% of the submission and no more than a 10% match to any one individual source can be directly quoted or closely paraphrased from sources, even if cited correctly. The similarity report that is provided when you submit your task can be used as a guide.

You must use the rubric to direct the creation of your submission because it provides detailed criteria that will be used to evaluate your work. Each requirement below may be evaluated by more than one rubric aspect. The rubric aspect titles may contain hyperlinks to relevant portions of the course.

*Tasks may **not** be submitted as cloud links, such as links to Google Docs, Google Slides, OneDrive, etc., unless specified in the task requirements. All other submissions must be file types that are uploaded and submitted as attachments (e.g., .docx, .pdf, .ppt).*

Part 1: Design Document

- A. Select one of the provided scenarios and complete the following:
 - 1. Describe a business problem that can be solved with a database solution and is in alignment with the chosen scenario.
 - 2. Justify why a NoSQL database solution will solve the identified business problem.
 - 3. Identify a NoSQL database type to solve the identified business problem.
 - 4. Explain how the business data will be used within the database solution.
- B. Discuss how the proposed database design addresses scalability concerns, including strategies that align with the chosen scenario.
- C. Outline the privacy and security measures that should be implemented in the proposed database design.

Part 2: Implementation

Note: The data files for each scenario are located in a folder titled "D597 Datasets" on the desktop of the WGU Virtual Lab environment. Be sure to pull the files from "Task 2" that relate to your chosen scenario.

Note: Submit your screenshots from the WGU Virtual Lab for each prompt with your design document.

- D. Implement the proposed database design in the WGU Virtual Lab environment by completing the following:
 - 1. Write script to create a database instance named "D597 Task 2" using the appropriate query language, based on your design in Part 1. Provide a screenshot showing the script and the

- database instance in the platform.
2. Write script to insert or map the data records from the chosen scenario JSON files into the database instance. Provide a screenshot showing the script and the data correctly inserted or mapped into the database.
 3. Write script for **three** queries to retrieve specific information from the database that will help to solve the identified business problem. Provide a screenshot showing the script for each query and each query successfully executed.
 4. Apply optimization techniques to improve the run time of your queries from part D3, providing output results via a screenshot.

Part 3: Presentation

Note: The audiovisual recording should feature you visibly presenting the material (i.e., not in voiceover or embedded video) and should simultaneously capture both you and your multimedia presentation.

Note: For instructions on how to access and use Panopto, use the "Panopto How-To Videos" web link provided below. To access Panopto's website, navigate to the web link titled "Panopto Access" and then choose to log in using the "WGU" option. If prompted, log in using your WGU student portal credentials, and then it will forward you to Panopto's website.

To submit your recording, upload it to the Panopto drop box titled "Task 2: Non-Relational Database Design and Implementation – MKN1 | D597." Once the recording has been uploaded and processed in Panopto's system, retrieve the URL of the recording from Panopto and copy and paste it into the Links option. Upload the remaining task requirements using the Attachments option.

- E. Present your functional database solution in the lab environment by doing the following:
1. Record a walk-through of your program appropriate for an audience of a project team with technical knowledge using Panopto. Record yourself describing your program. Your recording should capture both you and your functioning program. Your presentation should also demonstrate appropriate communication skills for your audience, including a professional appearance.
 2. Demonstrate the following in your recorded walk-through:
 - Discuss how database design and indexing strategy optimize performance.
 - Describe the technical environment used in your database implementation.
 - Demonstrate the functionality of the script in the lab environment.
 - Discuss how the script solves the identified business problem.
- F. Acknowledge sources, using in-text citations and references, for content that is quoted, paraphrased, or summarized.
- G. Demonstrate professional communication in the content and presentation of your submission.

File Restrictions

File name may contain only letters, numbers, spaces, and these symbols: ! - _ . * ' ()

File size limit: 200 MB

File types allowed: doc, docx, rtf, xls, xlsx, ppt, pptx, odt, pdf, csv, txt, qt, mov, mpg, avi, mp3, wav, mp4, wma, flv, asf, mpeg, wmv, m4v, svg, tif, tiff, jpeg, jpg, gif, png, zip, rar, tar, 7z

RUBRIC

A1:BUSINESS PROBLEM

NOT EVIDENT

The submission does not describe a business problem that can be solved with a database solution.

APPROACHING COMPETENCE

The submission describes a business problem that can be solved with a database solution, but the business problem is not appropriate for the chosen scenario or is not in alignment with the scenario.

COMPETENT

The submission describes a business problem that can be solved with a database solution and is in alignment with the chosen scenario. The business problem is appropriate for the scenario.

A2:DATABASE JUSTIFICATION

NOT EVIDENT

A justification is not provided.

APPROACHING COMPETENCE

The submission does not address why the NoSQL database solution will solve the identified business problem, or the justification is illogical or is not in alignment with the identified business problem.

COMPETENT

The submission logically justifies why a NoSQL database solution will solve the identified business problem.

A3:DATABASE TYPE

NOT EVIDENT

A NoSQL database type is not identified.

APPROACHING COMPETENCE

The submission identifies a NoSQL database type, but the identified NoSQL database type is not appropriate for the identified business problem.

COMPETENT

The submission identifies an appropriate NoSQL database type to solve the identified business problem.

A4:DATA USAGE

NOT EVIDENT

The submission does not explain how the business data will be used.

APPROACHING COMPETENCE

The submission explains how the business data will be used within the database solution, but the explanation contains in-

COMPETENT

The submission accurately explains how the business data will be used within the database solution.

accuracies or is not in alignment with the scenario.

B:SCALABILITY**NOT EVIDENT**

A discussion of how the proposed database design addresses scalability concerns is not provided.

APPROACHING COMPETENCE

The submission discusses how the proposed database design addresses scalability concerns but does not include strategies that align with the chosen scenario, or the discussion is incomplete.

COMPETENT

The submission discusses how the proposed database design addresses scalability concerns and includes strategies that align with the chosen scenario.

C:PRIVACY AND SECURITY**NOT EVIDENT**

An outline of the privacy and security measures that should be implemented is not provided.

APPROACHING COMPETENCE

The submission outlines the privacy and security measures that should be implemented in the proposed database design, but the measures provided contain inaccuracies or are not in alignment with the proposed database design.

COMPETENT

The submission accurately outlines the privacy and security measures that should be implemented in the proposed database design.

D1:DATABASE INSTANCE**NOT EVIDENT**

A script to create a database instance is not provided.

APPROACHING COMPETENCE

The script to create a database instance using query language is functional, but the script is not based on the logical data model created in part B, or the query language used is not appropriate. A screenshot showing the script or of the database instance created in the platform is not provided.

COMPETENT

The script to create a database instance using the appropriate query language is functional and is based on the logical database model created in part B. A clear screenshot showing the script and the database instance in the platform is provided.

D2:INSERT RECORDS

NOT EVIDENT

A script to insert or map the data records is not provided or is not functional.

APPROACHING COMPETENCE

The script of a statement to insert or map the data records from the chosen scenario JSON files into the database is functional, but the submission does not provide a screenshot showing the script or the data correctly inserted into the database.

COMPETENT

The script provided to insert or map the data records from the chosen scenario JSON files into the database is functional. A clear screenshot showing the script and the data correctly inserted or mapped into the database is provided.

D3:QUERIES**NOT EVIDENT**

Scripts of 3 or fewer queries to retrieve specific information from the database are not functional or not provided.

APPROACHING COMPETENCE

The scripts provided for 3 queries to retrieve specific information from the database are functional, but one or more of the queries will not help solve the identified business problem or the submission does not provide a screenshot showing the script for each query successfully executed.

COMPETENT

The scripts provided for 3 queries to retrieve specific information from the database that will help to solve the identified business problem are functional. A clear screenshot showing the script for each query successfully executed are provided.

D4:OPTIMIZATION**NOT EVIDENT**

The submission does not provide a screenshot of the output results demonstrating the optimization techniques to improve the run time of the queries.

APPROACHING COMPETENCE

The submission applies optimization techniques to improve the run time of some but not all of the queries written in part D3, or the optimization techniques do not improve the run time, or the submission does not provide the output results via a screenshot.

COMPETENT

The submission applies optimization techniques to improve the run time of each the queries written in part D3 and provides the output results via a screenshot.

E1:PRESENTATION: ORGANIZATION AND PROFESSIONALISM

NOT EVIDENT

A presentation is not submitted, or the video does not capture both the presenter and the functioning program for the duration of the presentation.

APPROACHING COMPETENCE

The presentation of program walk-through is unclear, is disorganized, or does not demonstrate appropriate communication skills. Or the presenter's appearance is unprofessional. Or the presentation does not include technical detail or is inappropriate for an audience of a project team with technical knowledge.

COMPETENT

The presentation of the program walk-through is clear, is organized, and demonstrates appropriate communication skills. For the duration of the presentation, the video captures *both* the presenter and the functioning program in a Panopto video recording. The presenter's appearance is professional. The presentation includes technical detail appropriate for an audience of a project team with technical knowledge.

E2:PRESENTATION: CONTENT**NOT EVIDENT**

A presentation depicting the program is not submitted, or the presentation does not reflect a functioning program.

APPROACHING COMPETENCE

The presentation does not demonstrate each of the given requirements, or 1 or more of the requirements are incomplete, or the information provided contains errors.

COMPETENT

The presentation thoroughly and accurately demonstrates each of the given requirements and reflects a functioning database solution.

F:SOURCES**NOT EVIDENT**

The submission does not include both in-text citations and a reference list for sources that are quoted, paraphrased, or summarized.

APPROACHING COMPETENCE

The submission includes in-text citations for sources that are quoted, paraphrased, or summarized and a reference list; however, the citations or reference list is incomplete or inaccurate.

COMPETENT

The submission includes in-text citations for sources that are properly quoted, paraphrased, or summarized and a reference list that accurately identifies the author, date, title, and source location as available, or the submission states no sources were used.

G:PROFESSIONAL COMMUNICATION**NOT EVIDENT****APPROACHING****COMPETENT**

Content is unstructured, is disjointed, or contains pervasive errors in mechanics, usage, or grammar. Vocabulary or tone is unprofessional or distracts from the topic.

COMPETENCE

Content is poorly organized, is difficult to follow, or contains errors in mechanics, usage, or grammar that cause confusion. Terminology is misused or ineffective.

Content reflects attention to detail, is organized, and focuses on the main ideas as prescribed in the task or chosen by the candidate. Terminology is pertinent, is used correctly, and effectively conveys the intended meaning. Mechanics, usage, and grammar promote accurate interpretation and understanding.

WEB LINKS

[Panopto Access](#)

Sign in using the "WGU" option. If prompted, log in with your WGU student portal credentials, which should forward you to Panopto's website. If you have any problems accessing Panopto, please contact Assessment Services at assessmentservices@wgu.edu. It may take up to two business days to receive your WGU Panopto recording permissions once you have begun the course.

[Panopto FAQs](#)

[Panopto How-To Videos](#)

[WGU Virtual Lab Environment](#)

SUPPORTING DOCUMENTS

[Task 2 Scenario 1.zip](#)

[Task 2 Scenario 2.zip](#)